

Tarun Gupta

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EDUCATION

Indian Institute of Technology (IIT) Indore, India
B.Tech., Computer Science and Engineering, 2018–2022
CGPA: 9.81/10 Dept. Rank 1, Institute Rank 2

PUBLICATIONS

[C1] All That Glitters is Not Novel: Plagiarism in AI Generated Research. **Gupta, T.**, Pruthi, D. ACL 2025. **Outstanding paper award (top 1% of accepted papers).**

[P1] Exploring the Limits of World Models. **Gupta, T.**, Pruthi, D. Preprint 2025. [\[Link\]](#)

[C2] Estimation of Speaker Age and Height from Speech Signal using Mixture-of-Experts Transformer Model. **Gupta, T.**, Truong, T.D., Anh, T.T., Chng, E.S. Inter-speech 2022.

EXPERIENCE

Research Associate, Indian Institute of Science Jul 2024 – Present

- Investigated plagiarism in AI research agents and discovered that 24% of AI-generated research documents plagiarize existing work without attribution, evading current detection systems [C1].
- Examined the limitations of “world models” as a framework for characterizing human-level understanding through case studies inspired from philosophy of science and epistemology [P1].
- [Exploratory; unsuccessful] Attempted to model synaptogenesis and pruning via filopodia within mathematical brain model NEMO [Dabagia et al., 2024], inspired by recent findings on its unexpected prevalence in adult brains [MIT News, 2022].
- [Exploratory; unsuccessful] Investigated whether co-reference chain heads’ concentration in document beginnings causes primacy bias in LLM attention patterns, contributing to the lost-in-the-middle problem [Liu et al., 2023].

AI Engineering Intern, REInt AI Jun 2024

- Implemented and tested spatio-temporal prediction models (ConvLSTM, SimVP) on real-world satellite data, predicting frames that can be used by weather prediction models like MetNet-2 in energy trading applications.

Senior Member Technical, D. E. Shaw (Hedge Fund) Jul 2022 – Jun 2024

- Developed High-Performance Computing (HPC) job scheduler for large-scale data processing; project [featured in Business Insider](#).
- Served as primary full-stack developer for LLM-powered employee review assistance software.
- Promoted to Senior Member Technical (L2) in 1.5 years (awarded to less than 10% of eligible staff).

Research Intern, NTU Singapore & Skit AI Dec 2021 – Jun 2022

- Built a mixture-of-experts (MoE) transformer model for speaker profiling, leading to a relative improvement of 18.5% in male age estimation and 8.6% in female age estimation on the TIMIT dataset [C2].

Research Intern, Computational Biology Dept., CMU Jan 2021 – Apr 2021

- Learned self-supervised computer vision techniques (SimCLR, MoCo, SwAV) and applied them to cryo-electron tomography data.

Research Intern, Mathematics Dept., IIT Indore Jan 2020 – Dec 2020

- Studied non-convex optimization and estimation theory.

ACCOLADES	Outstanding paper award, ACL 2025 (top 1% of accepted papers). [Link] Best undergraduate researcher award, IIT Indore, 2021. Department rank 1, institute rank 2 at IIT Indore. Silver medal, Inter IIT Tech Meet 9.0 (Bridge2i NLP Competition), 2021. Bronze medal, Inter IIT Tech Meet 8.0 (BITGRIT Data Science), 2020. JEE Advanced 2018: All India Rank 1055 (top 0.7%).
RELEVANT COURSEWORK	Computer Science: Computer Vision, Cryptography & Network Security, Advanced Algorithms, Machine Learning, Compiler Techniques, Computational Intelligence, Computer Networks, Computer Graphics & Visualization, Optimization Algorithms & Techniques, Operating Systems, Computer Architecture, Parallel Computing, Design & Analysis of Algorithms, Software Engineering, Automata Theory & Logic, Logic Design, Data Structures & Algorithms, Database & Information Systems[†], Discrete Mathematical Structures. Mathematics: Numerical Methods[†], Complex Analysis & Differential Equations II, Linear Algebra & Differential Equations I, Calculus. Other: Psychology[†], Physics[†], Chemistry[†]. AI for Medicine (Coursera): AI For Medical Diagnosis, AI For Medical Prognosis & AI For Medical Treatment. [Link] Quantum Computation (Online). [Link] ([†] indicates distinction in the course.)
TEACHING EXPERIENCE	Teaching Assistant, Indian Institute of Science Jan–Apr 2025 - Supported the Introduction to NLP (DS-207) course by conducting tutorial sessions, grading assignments, and providing one-on-one assistance to students. Head, Cynaptics Club (ML Student Club, IIT Indore) Aug 2020 – Jun 2021 - Organized tutorials and competitions, and mentored student projects in machine learning.